MIL-W-80046C
25 October 1985
SUPERSEDING
MIL-W-80046B
10 March 1977

MILITARY SPECIFICATION

WELDING MACHINES, RESISTANCE, STORED ENERGY, CAPACITOR DISCHARGE, DUAL RANGE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electrical resistance type welding machines which utilize stored energy capacitive discharge power for precision welding with equipment and accessories specified herein.

1.2 Classification. Welding machines shall be of the following classes and sizes. The class and size to be furnished shall be as specified (see 6.2.1).

Class A - Automatic and semi-automatic machines.
Class B - Manual and intermittent duty machines.

Size 1	-	3.6 watt-second capacity.
Size 2	-	15 watt-second capacity.
Size 3	-	50 watt-second capacity.
Size 4	-	100 watt-second capacity.
Size 5	-	200 watt-second capacity.
Size 6	-	500 watt-second capacity.
Size 7	-	1000 watt-second capacity.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3432

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-T-27D - Transformer and Inductor (Audio, Power, and High Power Pulse), General Specifications for.

MIL-W-45562 - Welding and Soldering Equipment, Supplies, and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-1276 - Leads for Electronic Component Parts.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in

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the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

RESISTANCE WELDER MANUFACTURERS ASSOCIATION (RWMA)

RWMA Bulletin Number 16 - Resistance Welding Equipment Standards.

(Application for copies should be addressed to the Resistance Welder Manufacturers Association, 1900 Arch Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The welding machine shall be new and one of the manufacturer's current model capable of welding fine wire and ribbon conductors in components of electrical equipment in accordance with the requirements herein. The class A machine shall be capable of semi-automatic and automatic operation. The class B machine shall be capable of performing manual and intermittent welding. The machine shall be designed and constructed so that it can be shipped fully assembled and ready for use when connected to the specified source of electrical and air supply. The machine shall include

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all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machines. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibration. Unless otherwise specified, either the U.S. Customary System of Units (US) or the the International System of Units (SI) shall be used to graduate measuring and indicating devices such as pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The welding machine and its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the welding machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

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3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The machine shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by federal, state, or local statutes in effect at point of installation (see 6.4).

3.2.9 Lubrication. When applicable, means shall be provided to insure adequate lubrication for all moving parts. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The welding machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the welding machine shall be painted in accordance with the manufacturer's commercial practice and color.

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3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28.

3.4 Components. The machine shall consist of not less than an enclosure, a power supply unit, a welding head, an electrical system, and electrodes and electrode holders.

3.4.1 Enclosure. The metal enclosure shall be constructed of corrosion-resistant material or treated to resist corrosion and shall have provisions for inspection and maintenance. Ventilation and cooling shall be adequate to prevent temperatures in excess of design requirements under normal operating conditions at rated power output. Where ventilation is provided by filtered forced air, the air inlet shall be provided with replaceable filters. The sizes 6 and 7 welding machine power supply and controls may be enclosed in two separate units.

3.4.2 Power supply unit. The power supply unit shall include the voltage regulator, capacitors, controls, lockout circuits and overvoltage protection devices. The primary line voltage shall pass through a rectifying circuit to the capacitors.

3.4.2.1 Voltage regulator. The voltage regulator shall maintain the capacitor charge voltage within ±1 percent with a line voltage fluctuation of ±10 percent.

3.4.2.2 Capacitors. The capacitors shall operate satisfactorily and maintain their rated microfared value when operated at ambient temperature. Above ambient and up to 104 degrees Fahrenheit the capacitors shall maintain their rated microfarad value within ±10 percent.

3.4.2.3 Charging lockout circuit. When specified (see 6.2.1), the power supply unit shall have the means to prevent welding initiation until the capacitors are charged to within 1 percent of that selected on the weld energy range control.

3.4.2.4 Discharge lockout circuit. The power supply unit shall have the means to prevent capacitors from being recharged until after they are discharged after a welding cycle.

3.4.2.5 Overvoltage protection. The power supply unit shall have an overvoltage protection device to prevent damage due to overvoltage.

3.4.2.6 Controls. The controls shall be adjustable for making repetitive welds as specified herein without readjustments. The controls shall consist of at least a dual range control, with a dual range indicator to expand the lower range scale readings, a weld energy control switch and dial, a "weld/no weld" control or master switch, and a pilot light.

3.4.2.6.1 Dual range switch. The dual range switch shall provide the means to change from a low welding range to a high welding range.

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3.4.2.6.2 Weld energy range selector control. The weld energy range selector control shall operate to regulate the weld energy to the desired level in stepless adjustments. The control shall be marked in watt-seconds or percent of full rated watt-seconds. The power supply shall be equipped with a turn down circuit which allows the capacitor bank voltage to track the weld energy control when it is adjusted either up or down.

3.4.2.6.3 Weld pulse selector switch. When specified (see 6.2.1), the welding machine shall be equipped with a weld pulse selector switch for selecting two or more weld output pulse widths to the degree specified.

3.4.2.6.4 Automatic control adaptors. When specified (see 6.2.1), the welding machine shall have provisions and be equipped with the proper adaptors for connection to automatic controls such as timers, programmers, and NC equipment.

3.4.2.7 Transformer. The transformer insulation shall be not less than the requirements of MIL-T-27D, class R.

3.4.2.8 Main disconnect switch. The welding machine shall have a main disconnect switch or a circuit breaker that will serve as an "on-off" switch and as an overload protection device. Unless otherwise specified (see 6.2.1), a circuit breaker shall be furnished and shall be mounted on the front control panel. A pilot light with a red lens shall be provided to indicate when the power supply is energized. The incoming electric power shall be fused when a circuit breaker is not furnished. When used as a protection device only, the fuse may be located on or inside the power supply unit.

3.4.2.9 Power supply cord. The power supply unit shall be furnished with not less than five feet of three-conductor flexible electric cord that conforms to ANSI/NFPA 70, type SO. The cord shall terminate in an appropriate 3-pole grounding plug. The cord and plug shall be of the proper capacity, temperature, and voltage rating for the welding machine size specified. The cord shall have a strain relief connector where the cord enters the enclosure. The third wire shall be grounded to the frame of the unit.

3.4.2.10 Control circuit voltage. Unless otherwise specified (see 6.2.1), control circuit voltage shall be no greater than 24 volts.

3.4.3 Welding head. Unless otherwise specified, the welding head for class A welding machines shall be of the pneumatically operated type and the welding head for class B welding machines shall be of the manually operated type. When specified (see 6.2.1), the welding head for class A welding machines shall be of the electrically or hydraulically operated type and the class B welding machine shall be of the electrically, hydraulically, or pnuematically operated type.

3.4.3.1 Electrode force switch. The welding head shall have a switch to prevent the initiation of the welding cycle until the predetermined electrode force has been established.

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3.4.4 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 70. Each welding machine shall operate on a single phase, 60 Hz source, at either 115 or 230 volts. Unless otherwise specified (see 6.2.1), the sizes 1 through 6 welding machines shall be initially wired for operation on 115 volts and size 7 welding machine shall be initially wired for operation on 230 volts.

3.4.4.1 Solid state components. Solid state design shall be used throughout for electronic components.

3.4.5 Electrodes and electrode holders. Unless otherwise specified (see 6.2.1), the electrodes and electrode holders shall be made of an alloy conforming to the requirements of RWMA Bulletin Number 16. Unless otherwise specified, the electrode holders for the sizes 1 through 6 welding machines shall be furnished for use with 1/8 inch diameter electrodes. When specified (see 6.2.1), the sizes 4 through 6 welding machines shall be furnished with electrode holders for use with 1/4 inch diameter electrodes. The size 7 electrode holders shall be for use with 1/4 inch diameter electrodes. The quantity, type, and size of electrodes required shall be as specified (see 6.2.1).

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the welding machine shall be not less than those specified in table I for the welding machine size specified.

3.6 Performance. The welding machine shall be capable of producing the watt-second weld energy range specified in table I and produce electrical resistance welds on materials with dimensions and at the repetition rate specified in table I.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. Unless otherwise specified (see 6.2.1), all standard equipment normally furnished by the manufacturer, shall be furnished.

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.10 Marking on plates, charts, and decals. All words on plates, charts, and decals shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. When these methods are not practical, the instructions shall be applied directly on the item by environmentally protected decalcomania transfer, metal wrap-around tag, stencil, silk screen, or by any other method suitable for the use intended. Markings shall be as permanent as the

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TABLE I. Machine size and capacities.

Machine size	Watt-second weld energy capacity	Millisecond discharge time allowance	Intermittent repetition rate/minute 1 minute on 1 minute off	Material weld capacity 1/ (combined thickness in inches)	
				Crosswire	Sheet/foil ribbon
1	Low range 0 to 0.25 High range 0.25 to 3.6	 0.25 to 3 0.25 to 3	 150 120	 .003 .006	 .001 .002
2	Low range 0.02 to 3 High range 0.1 to 15	 0.25 to 3.5 0.25 to 3.5	 120 100	 .0045 .018	 .0015 .005
3	Low range .25 to 10 High range .5 to 50	 0.25 to 4 0.25 to 6	 120 60	 .017 .035	 .004 .015
4	Low range 0.4 to 20 High range 0.7 to 100	 0.4 to 6 0.75 to 11	 100 60	 .019 .060	 .006 .023
5	Low range 2 to 30 High range 2 to 200	 0.25 to 4.5 1.5 to 12	 90 60	 .035 .070	 .012 .030
6	Low range 2 to 50 High range 10 to 500	 0.50 to 5 1.5 to 13	 80 35	 .035 .075	 .015 .040
7	Low range 4 to 100 High range 80 to 1000	 0.75 to 11 1.5 to 15	 60 20	 .060 .080	 .023 .050

1/ MIL-STD-1276, type N for chemical composition only.
NOTE: Requirements are not less than shown.
Where a range is shown, the requirements are from the smaller figure or less to the larger or more.

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normal life expectancy of the item to which they are applied and shall be capable of withstanding the environmental test and cleaning procedures specified for the item.

3.10.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each welding machine when applicable. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each welding machine. The nameplate shall contain the information listed below. If the welding machine is a special model, the model designation shall include the model of the basic standard and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.12 Workmanship. Workmanship of the welding machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. *QUALITY ASSURANCE PROVISIONS*

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in

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the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturers designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The welding machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The welding machine shall be operated at no load for not less than 15 minutes. Proper operation of all controls, motors adjusting mechanisms, and accessories shall be verified during the trial period.

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4.7.2 Welding test. Spot welds shall be made using the minimum and maximum size material in crosswire and sheet/foil specified in table I. Thirty consecutive welds shall be made in both size materials at the high and low weld energy capacity range and at the repetition rate specified in table I.

4.7.2.1 Weld inspection. Five weld specimens each of the high and low weld energy capacity range and at the minimum and maximum size material required for the sheet/foil test and the cross wire test shall be examined by 30 power magnification. There shall be no evidence of cracks or lack of fusion.

4.7.2.2 Weld strength test. After the welds have been inspected they shall be subjected to a strength test by pulling or peeling the two pieces apart. There shall be no separation of the weld nugget.

4.7.3 Performance test. The machine shall be adjusted to its maximum rated weld energy capacity and its rated repetition rate per minute and operated continuously for 2 hours of simulated welds. There shall be no overheating, arcing, or malfunctions of the machine.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Packaging, packing, and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or Level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The equipment covered in this specification is intended for use in contractor's plants, and military and government repair facilities. It is used for making connections in the circuits of semiconductors, integrated circuits, welding of wire and ribbons for transitor whiskers, strain gages, magnet wire, detonators, miniature electronic tubes, various thermocouple wire combinations, and hermetic sealing of semiconductors, rectifiers, crystal cases, and transformers.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class and size required (see 1.2).

c. First article, if required (see 3.1).

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d. If machine is required to be configured in a specific measure-system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.5 and 6.4).

g. Painting, if different (see 3.3.5).

h. Charging lockout circuit, if required (see 3.4.2.3).

i. Weld pulse selector switch, if required (see 3.4.2.6.3).

j. Automatic control adaptors, if required (see 3.4.2.6.4).

k. Main disconnect switch, if different (see 3.4.2.8).

l. Control voltage, if different (see 3.4.2.10).

m. Welding head, if different (see 3.4.3).

n. Electrical system, if different (see 3.4.4).

o. Electrical input power supply, if different (see 3.4.4).

p. Electrodes and electrode holders alloy, if different (see 3.4.5).

q. Size 4 through 6 electrode holder size, if different (see 3.4.5).

r. Electrodes, quantity, type, and size required (see 3.4.5).

s. Size and capacity, if different (see 3.5).

t. Additional performance features, if required (see 3.6.1).

u. Standard equipment, if different (see 3.7).

v. Specify and fully describe optional equipment, if required (see 3.8).

w. Specify applicable repair parts, if required (see 3.9).

x. Lubrication chart or plate, if required (see 3.10.1).

y. Nameplate, if different (see 3.10.2).

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z. First article inspection, if different (see 4.4).

aa. Quality conformance inspection, if different (see 4.5).

bb. If packaging, packing, and marking in accordance with MIL-W-45562 is required, specify level required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operators manuals, parts lists, wiring diagrams, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the welding machine itself. It does not require compliance with those OSHA requirements that concern the machine in its operating environment such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. OSHA limits the total hazard level of these hazards in the environment and does not limit the hazard level of individual machines in the environment. The procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The welding machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial machines supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required because the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1 (f).

6.5 Cross-reference of classification changes. Machine classification changes from the previous issue of this specification are as follows:

MIL-W-80046 & MIL-W-80046A	MIL-W-80046B	MIL-W-80046C
Class A - Machines with oil-filled capacitors	(No change)	Class A - Automatic and Semi-automatic machines
Class B - Machines with electrolytic capacitors	(No change)	Class B - Manual and intermittent duty machines

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MIL-W-80046 & MIL-W-80046A	MIL-W-80046B	MIL-W-80046C
Size 1 - 3.6 watt-second capacity	(No change)	(No change)
Size 2 - 15 watt-second capacity	(No change)	(No change)
	Size 3 - 50 watt-second capacity (New size)	(No change)
Size 3 - 100 watt-second capacity	Size 4 - 100 watt-second capacity	(No change)
Size 4 - 200 watt-second capacity	Size 5 - 200 watt-second capacity	(No change)
	Size 6 - 500 watt-second capacity (New size)	(No change)
Size 5 - 1,000 watt-second capacity	Size 7 - 1,000 watt-second capacity	(No change)

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Air Force - 99

Preparing activity:
DLA - IP

(Project 3432-0133)

Review activities:
Air Force - 84
DLA - GS

User activities:
Army - SM, AR
Navy - MC, AS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-80046C	Welding Machines, Resistance, Stored Energy

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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